

- 30** A radioactive source contains two species. One has a half-life of 4 days and decays by the emission of alpha particles whilst the other has a half-life of 3 days and emits beta particles. The initial count-rate is 352 min^{-1} , but when a sheet of paper is placed between the source and the detector this becomes 256 min^{-1} . The background count-rate is 16 min^{-1} .

What will be the count-rate after 12 days, without the paper present?

- A** 27 min^{-1} **B** 28 min^{-1} **C** 43 min^{-1} **D** 44 min^{-1}